

SHIVANSH GOEL

AI / ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer building production LLM applications with Retrieval-Augmented Generation (RAG), AI agents, and prompt engineering. Experienced integrating Google Gemini into live products, designing embedding and retrieval pipelines, engineering prompts for structured output, and hardening AI features with secure server-side model routing. Focused on accuracy, grounding, latency, and cost efficiency.

TECHNICAL SKILLS

LLM and Generative AI: LLM Integration, Google Gemini, AI Agents, Prompt Engineering, Structured Output, Profile Contexting

Retrieval and NLP: Retrieval-Augmented Generation (RAG), Semantic Retrieval, Embeddings, Document Chunking, TF-IDF, NLP, Sentiment Analysis

AI Security and Serving: AI Security Middleware, API-Key Proxying, Access-Key Authentication, LLM Request Gating, Redis Caching for Inference

Languages: Python, TypeScript, JavaScript (ES6+), SQL

Backend and Data: FastAPI, Node.js, REST APIs, Supabase, Redis, Cloudflare D1, JSON

Tools: Git, GitHub, Firebase Tools, Cloudflare Workers, VS Code, Debugging

PROFESSIONAL EXPERIENCE

Full Stack Developer, Intern

December 2023 – May 2025

Tedforge Solutions Private Limited

India

- Across a 17-month internship, built client-facing web applications with React and Angular and developed Node.js REST APIs, integrating data-driven, API-backed features.
- Improved MySQL and MongoDB data paths, reducing query overhead and making product interactions more responsive.
- Improved application stability by reducing reported bugs 40 percent through systematic debugging and refactoring, collaborating through Git and GitHub.

PROJECTS

Quizify – RAG-Based AI Assessment Platform [Live] | *Google Gemini, RAG, Embeddings, AI Agents, FastAPI, Redis, Supabase*

- Built an AI-powered assessment platform where AI agents ingest user-provided documentation and generate topic-based quizzes on demand, cutting manual quiz-creation effort significantly.
- Implemented a RAG pipeline that chunks and embeds source documents, then retrieves relevant context to ground Gemini's question generation and keep quizzes accurate and on-topic.
- Engineered prompts for structured, schema-consistent output (question, options, answer, explanation) and applied profile contexting to tailor difficulty and style per learner.
- Built an AI Security Middleware that proxies all Gemini calls server-side with no client-side API key, and used Redis caching to reduce repeat model calls, latency, and cost.

FiTrack AI – Adaptive AI Fitness Platform [Live] | *Google Gemini, NLP, Machine Learning, Python, Next.js*

- Built an AI-powered fitness platform that analyzes user performance patterns and adaptively adjusts workout intensity for optimal results.
- Integrated Gemini with profile contexting for personalized, goal-aware guidance, and applied NLP to turn raw performance data into plain-language, actionable feedback.

GhostRelay – Privacy-First Email Aliasing Platform [Live] [Source] | *Next.js, React 19, Cloudflare Workers, Cloudflare D1, Resend*

- Built a privacy-first email aliasing platform on Cloudflare's edge network that forwards to a user's real inbox while keeping the real address hidden from spam, breaches, and tracking.
- Implemented one-click alias creation with per-alias enable and disable controls, letting users instantly stop spam and pinpoint which service leaked their email.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

2023

COER University

Roorkee, India

- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering.

CERTIFICATIONS

Python 3.11 Programming with FastAPI Backend Framework | **TypeScript 5.2** – Microsoft

Angular 16 with TypeScript 5.2 and PrimeNG 16 | **Onsite Technical Internship – Trainee Software Developer (Full Stack)**